Algorithm 1 Forward workflow of MWAF-Net

Require: Preprocessed PCG segment $x_{\text{norm}} \in \mathbb{R}^{1\times L}$; wavelet set $\mathcal{W} = \{\text{morl}, \text{db4}, \text{sym4}, \text{coif4}, \text{Haar}, \text{mexh}\}$; current model parameters

Ensure: Class logits $\hat{y}$

1: **for** each wavelet template $b \in \mathcal{W}$ **do**
2: Apply input MSWC, BN, ReLU, max pooling, and dropout.
3: Extract features with two EW-TCN blocks; apply dropout and branch attention.
4: Obtain $F_b \in \mathbb{R}^{B\times 128\times 375}$ and apply branch-wise instance normalisation.
5: **end for**
6: Compute softmax-normalised global branch weights α_b.
7: Fuse wavelet-specific features by $F_{\text{fused}} = \sum_b \alpha_b F_{b,\text{norm}}$.
8: Apply fusion convolution, BN, ReLU, GCA, and dropout.
9: Apply a post-fusion EW-TCN block to obtain 256-channel features.
10: Apply global pooling and the classifier head to obtain $\hat{y}$.
11: **return** $\hat{y}$.

Abbreviations: PCG, phonocardiogram; MSWC, multi-scale wavelet convolution; EW-TCN, enhanced wavelet temporal convolutional network; GCA, gated coordinate attention. BN, batch normalisation; ReLU, rectified linear unit; B, batch size. Wavelet labels: morl, Morlet; db4, Daubechies 4; sym4, Symlet 4; coif4, Coiflet 4; mexh, Mexican Hat.